

Distribución	Densidad de Probabilidad	Θ_X	Parámetros	Esperanza y Varianza
Binomial	$\binom{n}{x} p^x (1-p)^{n-x}$	$x = 0, \dots, n$	n, p	$\mu_X = n p$ $\sigma_X^2 = n p (1-p)$ $M(t) = [p e^t + (1-p)]^n, \quad t \in \mathbb{R}$
Geométrica	$p (1-p)^{x-1}$	$x = 1, 2, \dots$	p	$\mu_X = 1/p$ $\sigma_X^2 = (1-p)/p^2$ $M(t) = p e^t / [1 - (1-p) e^t], \quad t < -\ln(1-p)$
Binomial-Negativa	$\binom{x-1}{r-1} p^r (1-p)^{x-r}$	$x = r, r+1, \dots$	r, p	$\mu_X = r/p$ $\sigma_X^2 = r(1-p)/p^2$ $M(t) = \left\{ p e^t / [1 - (1-p) e^t] \right\}^r, \quad t < -\ln(1-p)$
Poisson	$\frac{(\nu t)^x e^{-\nu t}}{x!}$	$x = 0, 1, \dots$	ν	$\mu_X = \nu t$ $\sigma_X^2 = \nu t$ $M(t) = \exp \left[\lambda (e^t - 1) \right], \quad t \in \mathbb{R}$
Exponencial	$\nu e^{-\nu x}$	$x \geq 0$	ν	$\mu_X = 1/\nu$ $\sigma_X^2 = 1/\nu^2$ $M(t) = \nu / (\nu - t), \quad t < \nu$
Gamma	$\frac{\nu^k}{\Gamma(k)} x^{k-1} e^{-\nu x}$	$x \geq 0$	k, ν	$\mu_X = k/\nu$ $\sigma_X^2 = k/\nu^2$ $M(t) = [\nu / (\nu - t)]^k, \quad t < \nu$
Normal	$\frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right]$	$-\infty < x < \infty$	μ, σ	$\mu_X = \mu$ $\sigma_X^2 = \sigma^2$ $M(t) = \exp(\mu t + \sigma^2 t^2 / 2), \quad t \in \mathbb{R}$
Log-Normal	$\frac{1}{\sqrt{2\pi}(\zeta x)} \exp \left[-\frac{1}{2} \left(\frac{\ln x - \lambda}{\zeta} \right)^2 \right]$	$x \geq 0$	λ, ζ	$\mu_X = \exp \left(\lambda + \frac{1}{2} \zeta^2 \right)$ $\sigma_X^2 = \mu_X^2 (e^{\zeta^2} - 1)$ $E(X^r) = e^{r\lambda} M_Z(r\zeta), \text{ con } Z \sim \text{Normal}(0,1)$
Uniforme	$\frac{1}{(b-a)}$	$a \leq x \leq b$	a, b	$\mu_X = (a+b)/2$ $\sigma_X^2 = (b-a)^2 / 12$ $M(t) = [e^t b - e^t a] / [t(b-a)], \quad t \in \mathbb{R}$
Beta	$\frac{1}{B(q, r)} \frac{(x-a)^{q-1} (b-x)^{r-1}}{(b-a)^{q+r-1}}$	$a \leq x \leq b$	q, r	$\mu_X = a + \frac{q}{q+r} (b-a)$ $\sigma_X^2 = \frac{q r (b-a)^2}{(q+r)^2 (q+r+1)}$
Hipergeométrica	$\frac{\binom{m}{x} \binom{N-m}{n-x}}{\binom{N}{n}}$	$\max\{0, n+m-N\} \leq x \leq \min\{n, m\}$	N, m, n	$\mu_X = n \frac{m}{N}$ $\sigma_X^2 = \left(\frac{N-n}{N-1} \right) n \frac{m}{N} \left(1 - \frac{m}{N} \right)$